Machine Learning and Computer Vision for Autonomous Vehicles

Permutation-Invariant Knowledge Distillation for Motion Prediction in Autonomous Vehicles

Where prediction sits in the self-driving pipeline

Perception	Prediction	Planning	Control
detect & track agents from sensors	forecast future trajectories	choose a safe ego trajectory	steering, throttle, brake

HiVT model

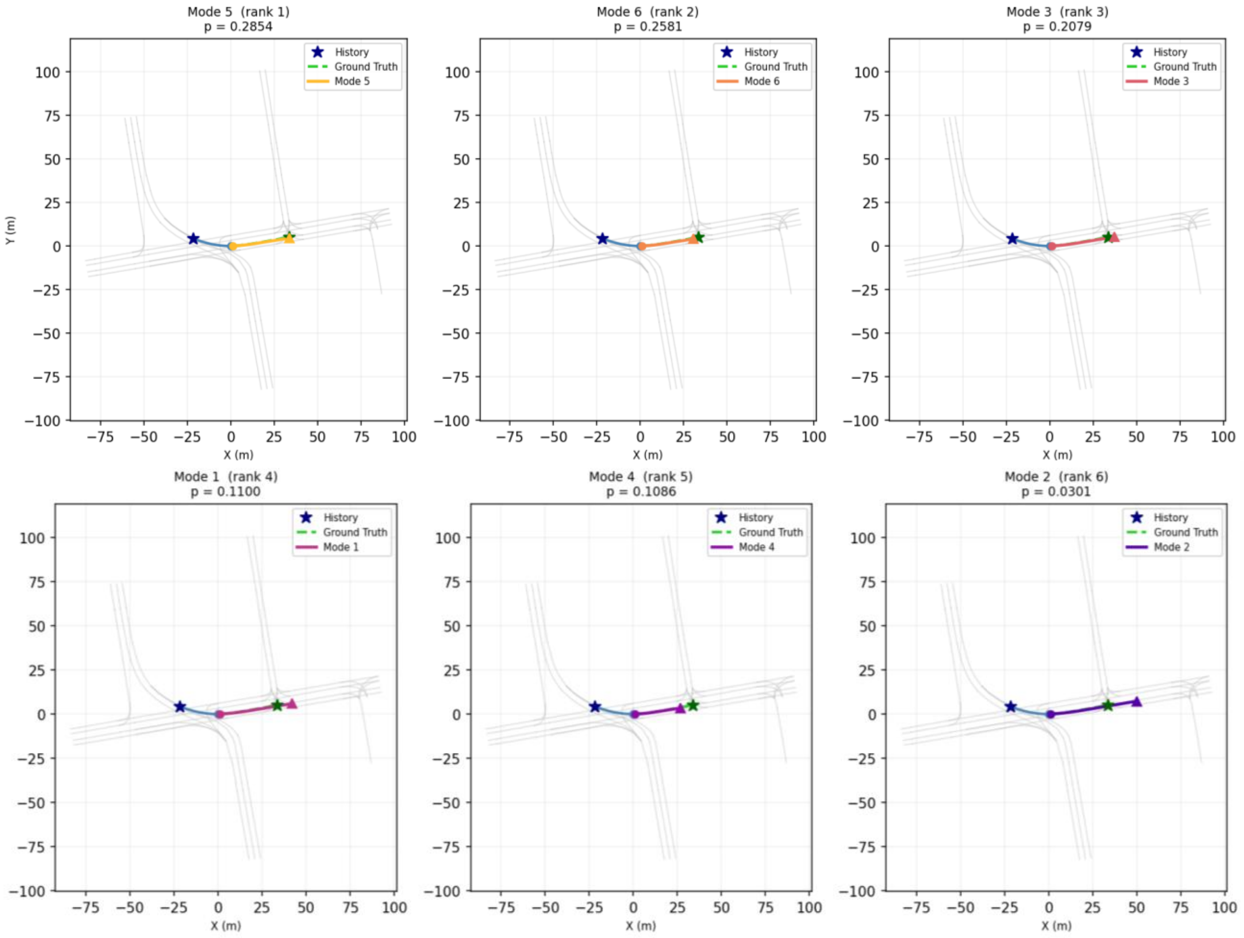
Why prediction is the hard middle

- Its output feeds the planner directly
- Modular pipeline pros: interpretable, testable, traceable

The output must be multi-modal

- A car at an intersection may turn OR go straight
- A single averaged path follows neither road — dangerous for planning

Motion prediction = multi-modal trajectory forecasting



Input → Output

- Input: 2 s of past agent states + an HD map (vectorised)
- Output for the focal agent: K = 6 future trajectories over 3 s, each with a probability π_k
- Argoverse 1: 10 Hz → 20 observed steps, H = 30 predicted steps

Evaluated best-of-K

- minADE / minFDE / MR: is ANY of the 6 modes close to the truth?
- brier-minFDE: also rewards high probability on the right mode
- … but best-of-K ignores whether the predicted uncertainty is honest — remember this

Modern predictors are accurate — and too heavy

Training side

- SOTA (MTR, MTR++, Wayformer, QCNet) is transformer-based and very accurate
- Trained on massive clusters with real + simulated fleet data

Deployment side

- On-vehicle: fixed latency window, strict power budget, 10–30 Hz
- Neither end is reproducible on one researcher's hardware

How small can a competitive predictor be, and can lost accuracy be recovered **without enlarging the model or worsening calibration?**

Metrics used

- minADE — mean L2 distance over all steps, best mode
- minFDE — L2 distance at the endpoint, best mode (primary)
- MR (miss rate) — fraction of scenes whose best endpoint is > 2 m off
- calibration error — gap between nominal and empirical coverage of the predicted intervals (how far the reliability curve strays from the diagonal)
- brier-minFDE = average(minFDE + $(1 - p)^2$)

The tool: knowledge distillation — and why HiVT

Knowledge distillation (KD)

The plan in one line

- Shrink HiVT by embedding width; find where accuracy falls off (the gap to close)
- Use HiVT-128 as teacher to recover that gap in a small student
- Keep the student architecturally identical to a plain small HiVT → zero added inference cost

Why HiVT?

- Rare strong model that trains AND runs on one GPU
- So teacher + students + the whole comparison fit on the available hardware
- Chosen over heavier MTR for exactly this reason

Why Argoverse 1?

- The dataset HiVT was natively designed for
- Waymo has no public checkpoint; Argo 2 / larger sets need bigger models + more data (scaling law)

HiVT: a hierarchical vector transformer

Local encoder (per agent)

Temporal attention — over the agent's own motion history
Agent–agent attention — neighbours within a local radius
Agent–map attention — nearby lane segments (vectors)

Global interaction

- models agent–agent interactions
- produces embedding per agent
- projects to 6 distinct mode

Laplace-mixture decoder

- maps each of the 6 mode embeddings to a trajectory
- per-timestep Laplace distribution

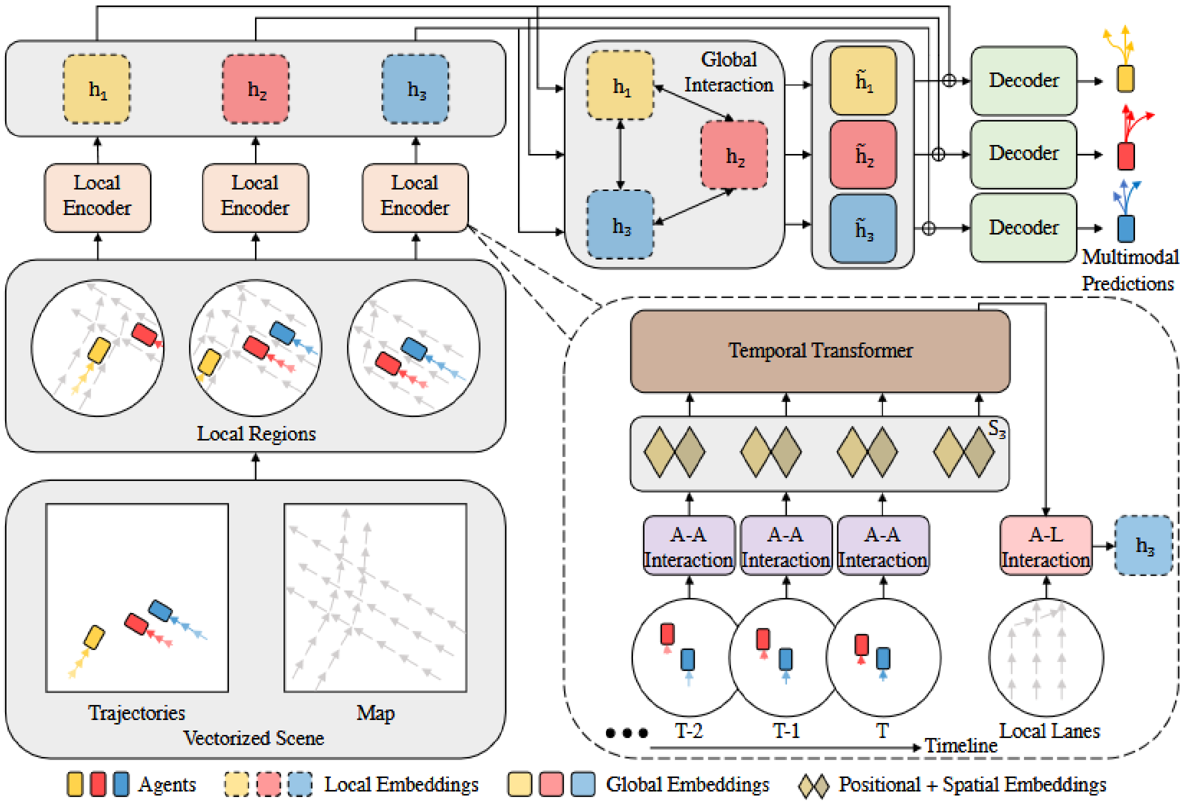


Figure 1. Overview of HiVT. A-A Interaction and A-L Interaction denote Agent-Agent and Agent-Lane Interaction respectively.

Figure from Zhou et al. (HiVT).

HiVT's output and training

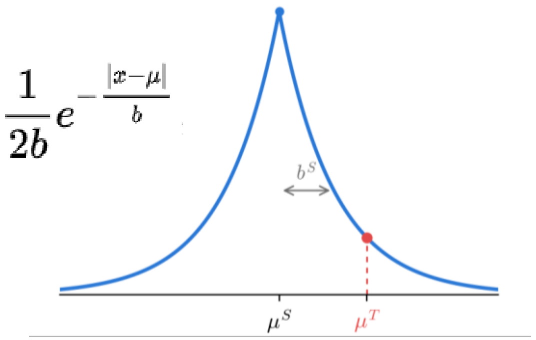
$$p(y) = \Sigma_k \ \pi_k \cdot \mathrm{Laplace}(\, y \mid \mu_k , b_k \,) \qquad k = 1 \ldots K,\ \ \pi_k \geq 0,\ \ \Sigma\, \pi_k = 1$$

each of the K=6 modes = one whole trajectory, modelled per-step as a factorised Laplace

Laplace mixture

μ_k	b_k	π_k
location	scale	weight

$$f(x \mid \mu, b) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}}$$



WTA loss

- error of all 6 modes vs. ground truth
- Regression loss only to winning trajectory
- different agents, different winning modes
- **modes specialize**

- classification loss on π

Which weights the WTA loss updates:

Winning mode's projection block in the global interactor (multihead_proj, the only per-mode weights) — updated per agent, for its winner only
Shared decoder (MLP head) updated by every winning trajectory
Shared backbone (local encoder + attention layers) — updated for all agents

Across a batch, agents choose different winners → all 6 mode blocks get trained, each from the subset of agents that selected it

RESULT 1

Is there a gap to close? From-scratch baselines

Model	Params	minFDE ↓	MR ↓	calib. err
HiVT-128 (teacher)	2.56 M	0.969	0.092	0.047
HiVT-64	653 k	1.030	0.103	0.042
HiVT-32 (student)	170 k	1.157	0.122	0.033
HiVT-16 (student)	46 k	1.605	0.199	0.028

Param count scales ≈ d^2 → each width-halving removes ~¾ of the parameters

Tuned best learning rate (cosine annealing), number of attention heads, batch size, workers, bf16 accuracy

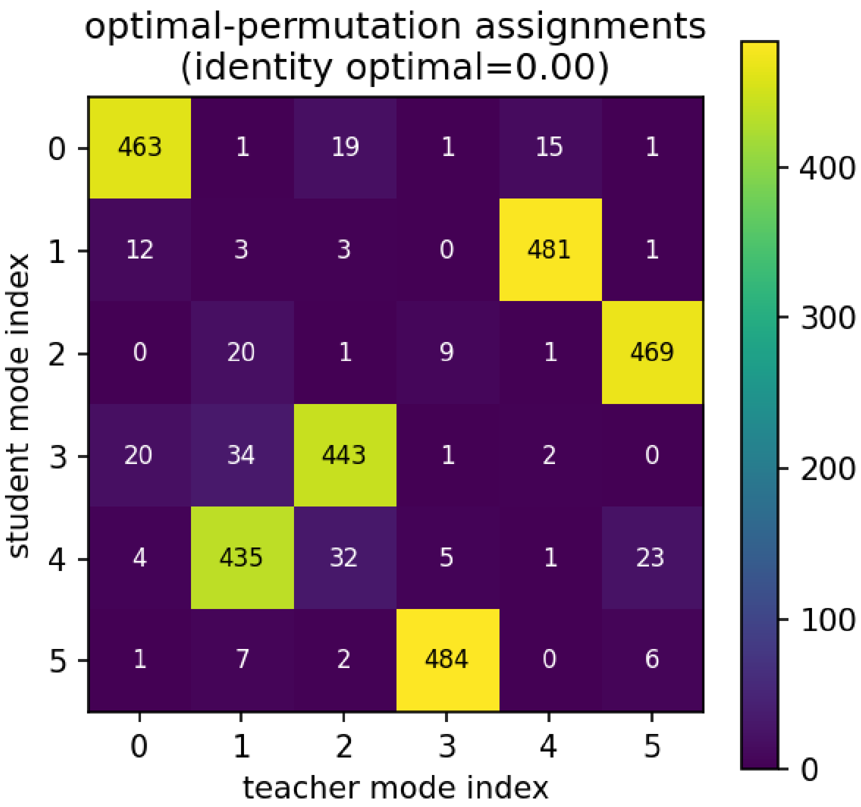
Accuracy falls faster-than-linearly

- 64→32 costs +0.13 m minFDE
- 32→16 costs +0.45 m — the penalty steepens
- So the smallest students have the MOST to recover

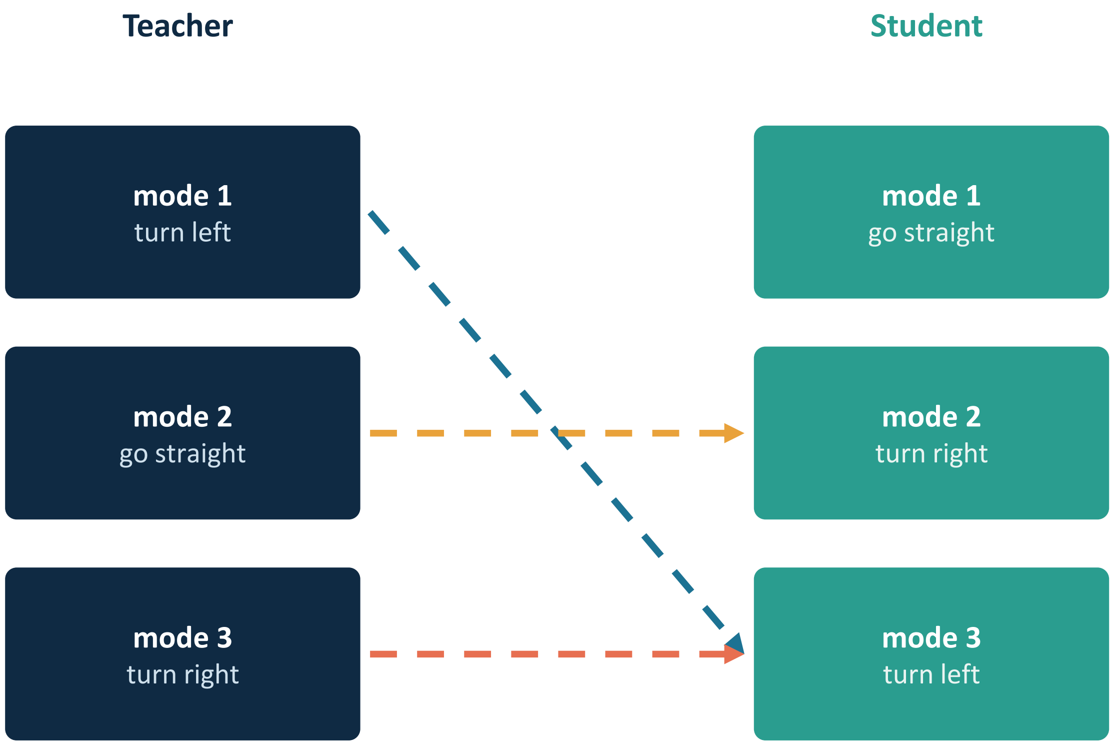
Calibration is already good

- calib. err ≤ 0.047 everywhere (lowest at the smallest widths)
- So calibration is NOT a pre-existing defect — a method that damages it is unacceptable

Why naive index-aligned distillation fails



The identity (index) pairing is the optimal assignment in **0% of scenes.**



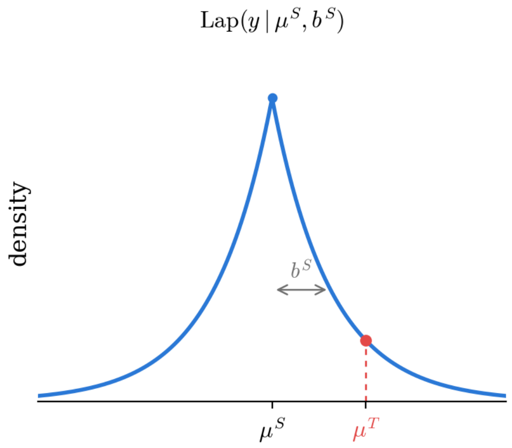
same behaviour lives in different slots → match by MEANING, not index

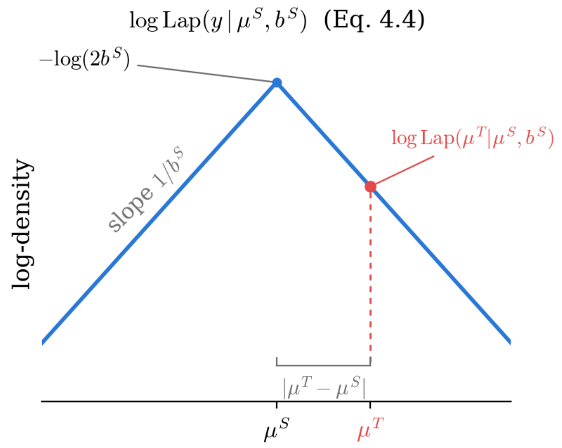
A permutation-invariant mixture objective

Score each teacher mode under the student's FULL mixture density — no matching at all.

$$L_{v1} = -\Sigma^f \pi^{fT} \cdot \log p_s(\mu^{fT}) = -\Sigma^f \pi^{fT} \log \Sigma_k \pi_k^s \,\text{Laplace}(\mu^{fT} \mid \mu_k^s, b_k^s)$$

teacher modes = a confidence-weighted SET of soft targets · scored under the student mixture p_s





1. For the focal agent
2. Per student mode K, per teacher mode F, Per time-step score
3. Trajectory: Combine time steps: p(x1)p(x2)…p(H) = p_0
4. Multiply by π (student mode probablity)
5. Score the teacher mode F: log (π_0*p_0 + π_1*p_1+ …+ π_5*p_5)
6. Score all the teacher modes multiplied by the teacher π

Updates all the weights

$$\mathcal{L}_{\mathrm{v1}} = -\sum_{f=1}^{K_T} \pi_f^T \log p_S(\boldsymbol{\mu}_f^T) = -\sum_{f=1}^{K_T} \pi_f^T \log \sum_{k=1}^{K_S} \pi_k^S \,\mathrm{Laplace}(\boldsymbol{\mu}_f^T \mid \boldsymbol{\mu}_k^S, \boldsymbol{b}_k^S)$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{HiVT}} + \lambda\, \mathcal{L}_{\mathrm{KD}}$$

Accuracy up — but calibration collapses

HiVT-32, A/B at λ = 0.5: the same recipe, distillation term switched on.

✓ Geometry improves

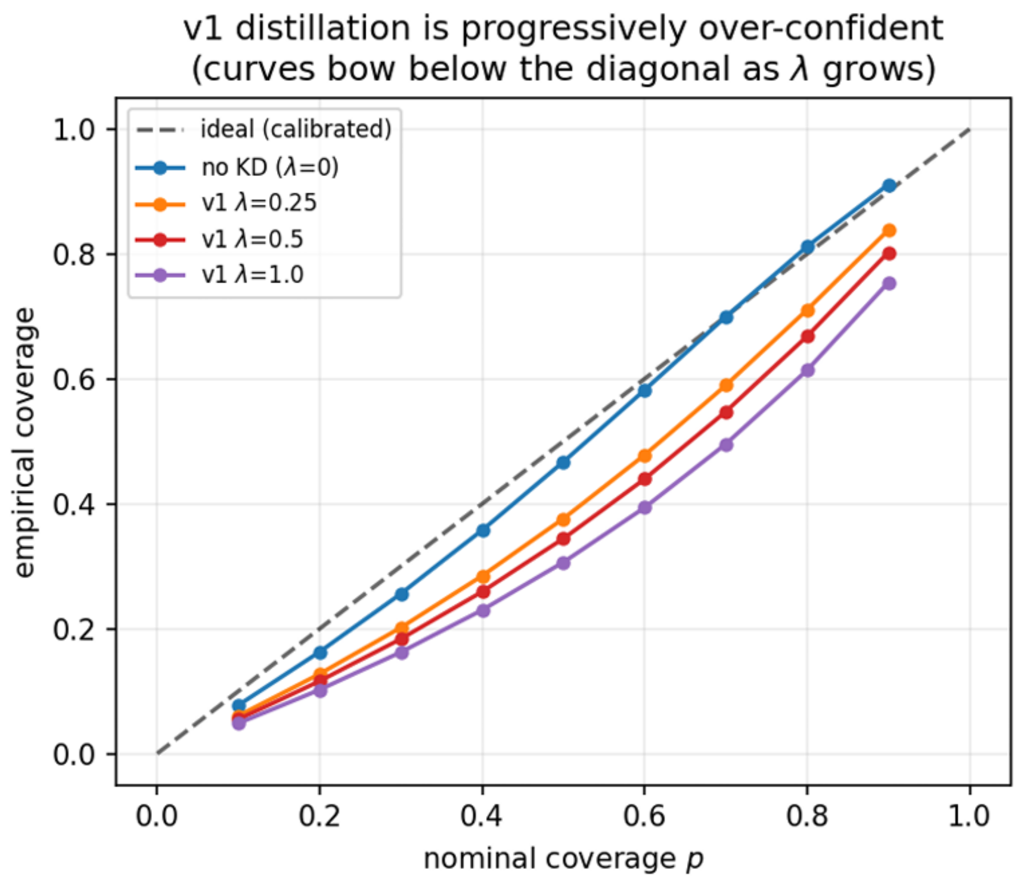
minFDE	1.157 → 1.050	**−9.2%**
MR	0.122 → 0.105	**−13.4%**
minADE	0.736 → 0.696	**−5.5%**
brier-minFDE	1.818 → 1.724	**−5.2%**

✗ Calibration collapses

mixNLL	26.6 → 38.0	**+43%**
calib. error	0.033 → 0.184	**× 5.6**
mean scale b	0.418 → 0.268	**−36%**
cov@p90	0.903 → 0.711	**under-covers**

The 90% predicted interval now captures only 71% of ground-truth points — the student is over-confident.

What calibration is — and why v1 breaks it



Calibration, in one line

- A p% predicted interval should contain the truth p% of the time
- Reliability curve below the diagonal = intervals too narrow = over-confident
- It matters because a downstream planner consumes the whole distribution, not just the best mode

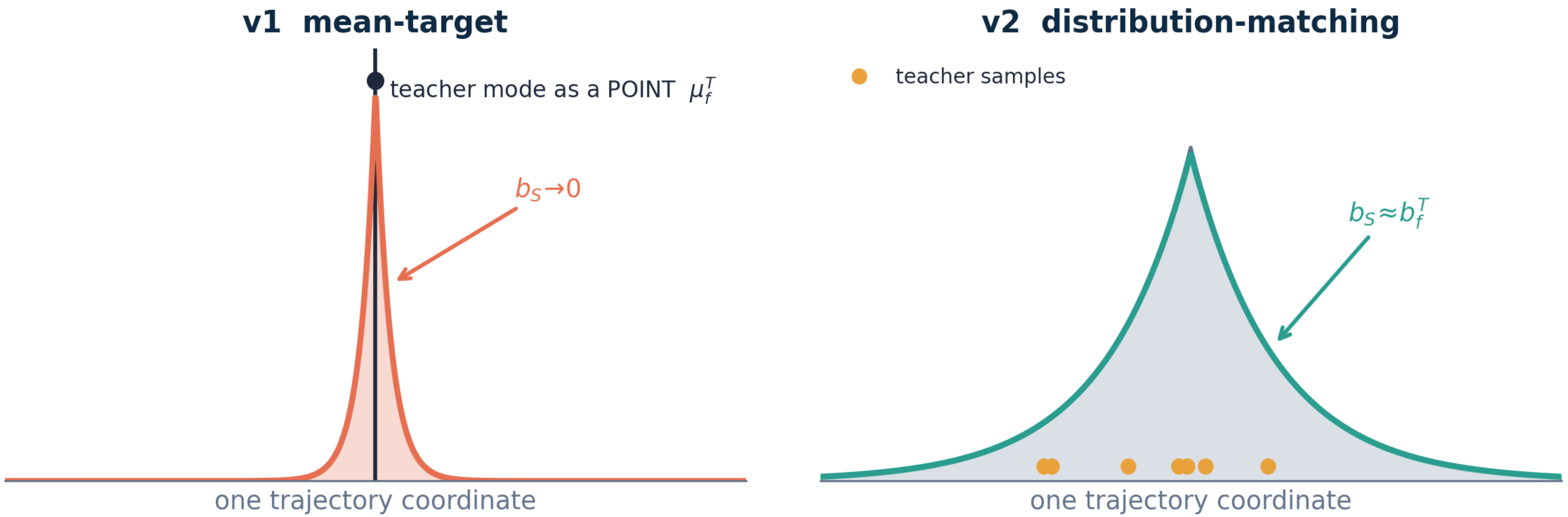
Why v1 breaks it (read off the Laplace NLL)

$$-\log \mathrm{Laplace}(\,\mu \mid \mu, b\,) = \log 2b$$

Scoring a POINT target is minimised as $b \to 0$. **Nothing opposes shrinkage.**

The baseline regression NLL keeps the residual term $|y-\mu|/b$, which grows as $b \to 0$ — so it balances at a finite, calibrated scale. v1 removed that anchor.

v2: score SAMPLES that carry the teacher's spread

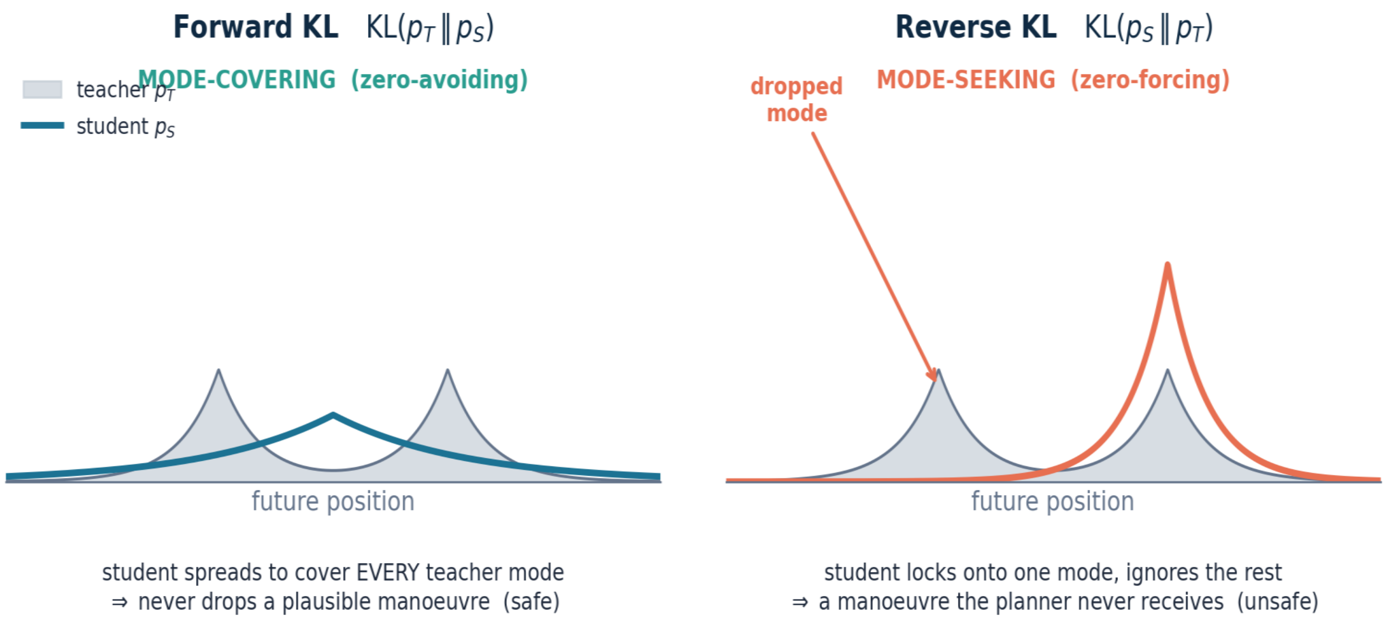


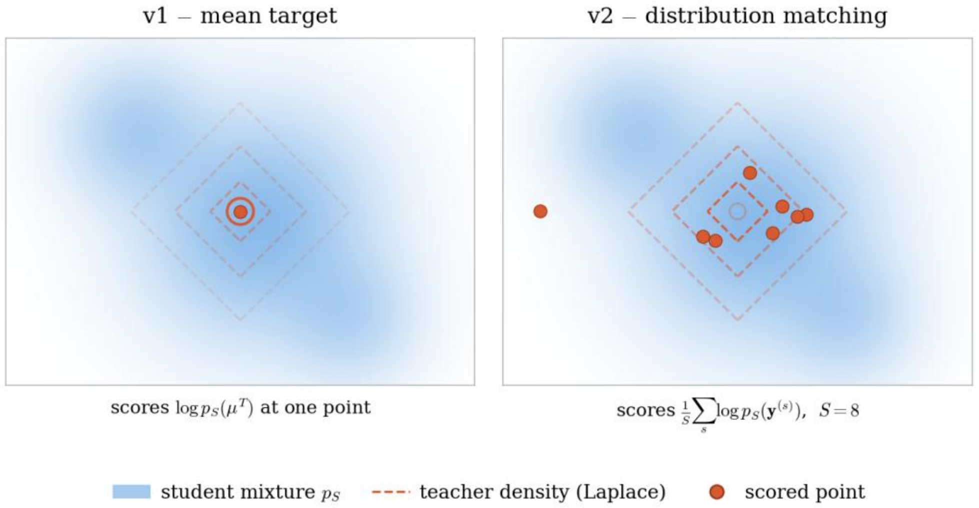
nothing opposes shrinkage: $-\log \mathcal{L}(\mu\,|\,\mu, b) = \log 2b \downarrow$ **as** $b \to 0$
student becomes OVER-CONFIDENT

samples carry the teacher's spread b_f^T $\Rightarrow$ **student must stay wide**
calibration is PRESERVED

v2 replaces the point target μ^{fT} by S = 8 Monte-Carlo samples drawn with the teacher's own scale b^{fT}. **A collapsed student misses the outer samples and is penalised → it must stay wide.**
Similar to Ensemble Distribution Distillation

v2 = forward Kullback–Leibler (KL) divergence





v2 = forward KL

$-\,E_{y\sim p_T}[\log p_S] = KL(p_T \| p_S) + H(p_T)$. Minimising v2 over the student IS minimising the forward KL.

Mode-covering is the safe direction

Dropping a plausible manoeuvre the planner never receives is the dangerous error. An extra low-probability mode is merely conservative.

Accuracy retained, calibration repaired

Metric	no KD	v1	v2	teacher
minFDE ↓	1.157	1.050	**1.050**	0.969
MR ↓	0.122	0.105	**0.106**	0.092
mixNLL ↓	26.6	38.0	**24.2**	21.1
calib. err ↓	0.033	0.184	**0.028**	0.047
mean scale b	0.418	0.268	**0.413**	0.358
cov@p90 (→0.90)	0.903	0.711	**0.909**	0.894

Full accuracy gain of distillation at zero calibration cost.

v2 delivers both

- Ties v1 on geometry EXACTLY (minFDE 1.0505 vs 1.0503)
- Calibration error 6.7× better than v1 — and below both the baseline AND the teacher
- Mean scale recovers to the student’s calibrated width (0.413)
- cov@p90 back to nominal (0.909)

Smaller student HiVT-16

Metric	no KD (n=3)	v2, λ=1.0 (n=3)	Δ
minADE	0.875 ± 0.021	0.771 ± 0.005	-11.8%
minFDE	1.562 ± 0.080	1.224 ± 0.018	-21.6%
MR	0.189 ± 0.016	0.130 ± 0.002	-31.0%
mixNLL	34.9 ± 1.0	29.4 ± 0.3	-15.7%
calib. err	0.0283 ± 0.0010	0.0229 ± 0.0004	-19.1%

−21.6%
minFDE over 3 seeds

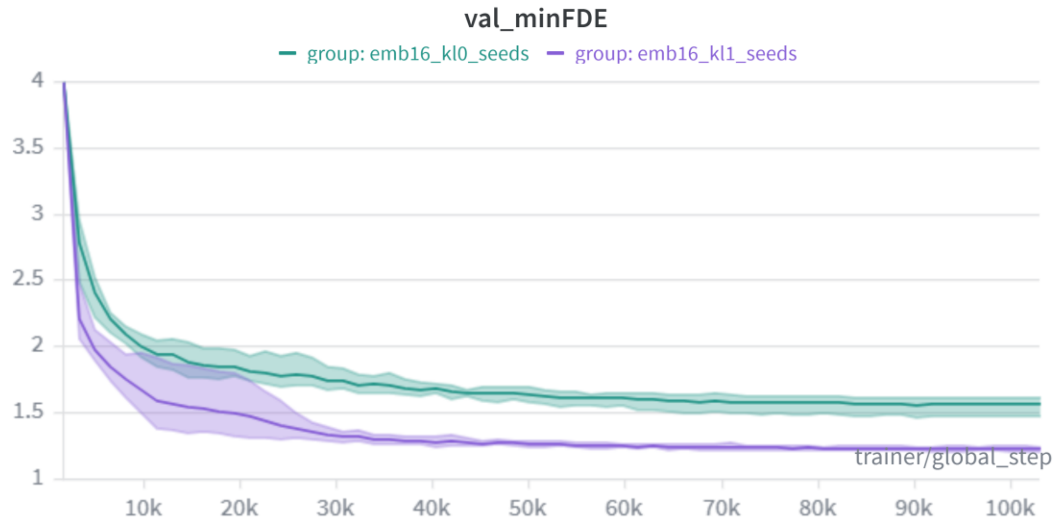
2.5×
larger relative gain than HiVT-32

Seed robustness

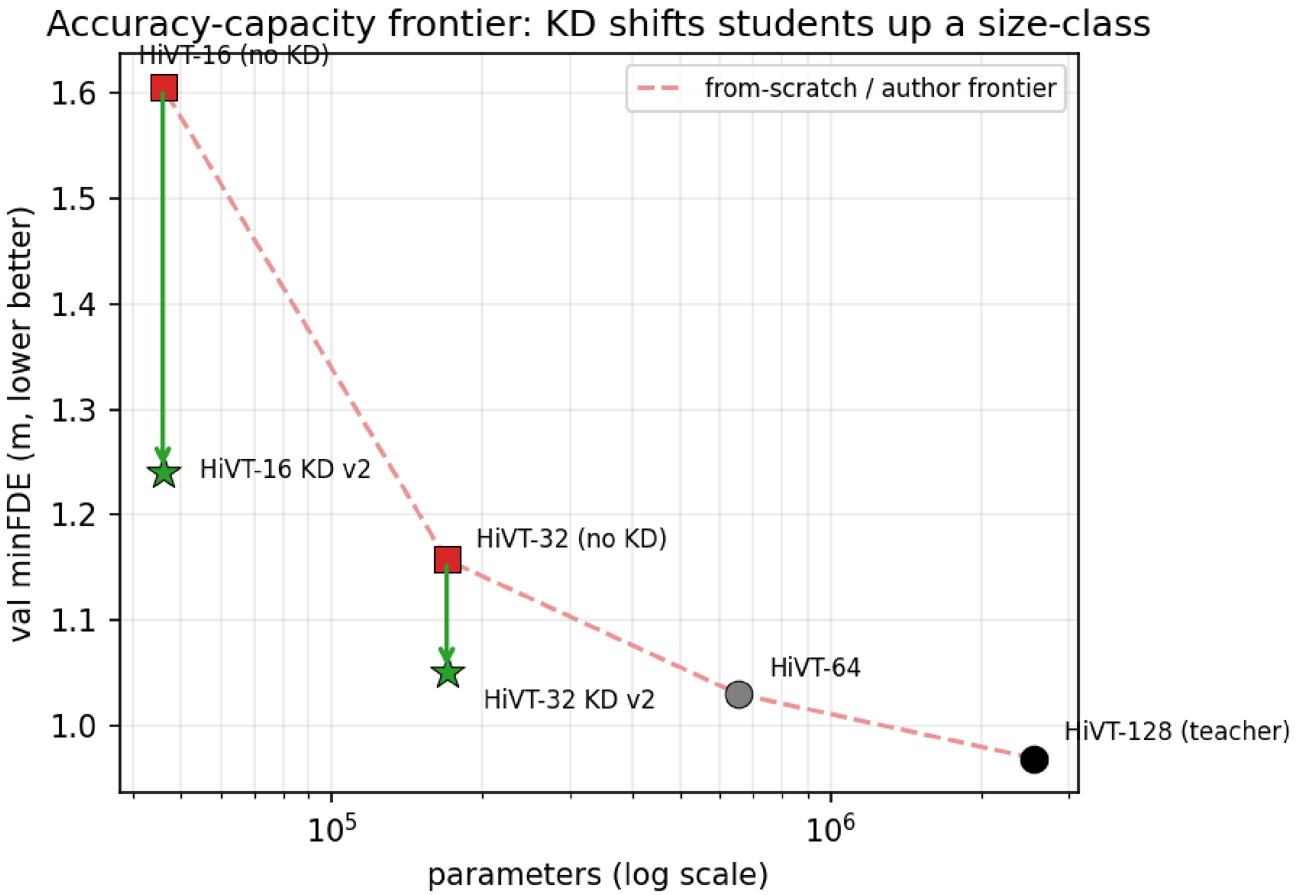
- 3 seeds per arm; the two seed clouds do NOT overlap on any single seed
- Worst v2 run (1.24) still beats the best no-KD run (1.47)
- Calibration also improves (calib. err −19%, mixNLL −16%) — no tax
- 1 of 4 from-scratch HiVT-16 seeds collapsed into a degenerate scale-inflated basin (b≈0.9) None of the distilled runs did

Tuning

- 2 attention heads
- 3 seeds for each



Distillation buys ~a full size-class — for free



Reading the frontier

- HiVT-32 KD (1.050) lands on top of from-scratch HiVT-64 (1.030) — a model with 3.8× more parameters
- Recovers ≈84% of the 32→64 gap at 6.6% of the teacher's parameters
- HiVT-16 KD recovers ≈81% of the 16→32 gap
- The student is architecturally unchanged → zero added inference cost
- KD helps the smaller student more because it has a larger gap to the teacher and too little capacity to learn good multi-modal structure from the sparse winner-takes-all signal alone

Metrics through time

	minADE			minFDE		
Model	1 s	2 s	3 s	1 s	2 s	3 s
HiVT-16 (no KD)	0.338	0.544	0.886	0.345	0.803	1.605
HiVT-16 KD v2	**0.317**	**0.500**	**0.776**	**0.318**	**0.682**	**1.240**
Δ vs. no KD	−6.2%	−8.1%	−12.4%	−7.7%	−15.1%	−22.7%
HiVT-32 (no KD)	0.305	0.478	0.736	0.299	0.642	1.157
HiVT-32 KD v2	**0.300**	**0.461**	**0.698**	**0.289**	**0.594**	**1.050**
Δ vs. no KD	−1.6%	−3.5%	−5.1%	−3.3%	−7.6%	−9.2%
HiVT-128 (teacher)	0.290	0.441	0.661	0.272	0.556	0.969

Memory, Latency

Memory

- Parameters: 15× smaller at width 32, 55× at width 16 (exact)
- Peak GPU memory: up to 6.3× less (HiVT-16 vs teacher at batch 128)
- Ratio WIDENS with batch size
- Enables larger batches, smaller GPUs, co-locating models

Batch	HiVT-128	HiVT-32	HiVT-16	T/S32	T/S16
1	70.4	40.5	36.6	1.7×	1.9×
8	203.6	76.8	56.7	2.7×	3.6×
32	799.2	241.4	147.9	3.3×	5.4×
64	1519.3	435.7	257.0	3.5×	5.9×
128	3409.8	954.3	543.5	3.6×	**6.3**×

Latency — fixed overhead

- CPU, online (bs=1): only ~3× despite 15× fewer params — a fixed graph-construction / dispatch floor dominates
- CPU, modest batching: rises to ~5.5× (HiVT-32), ~11× (HiVT-16)
- GPU: advantage nearly vanishes (~1–2.6×) — small matmuls never saturate it

	CPU latency (ms)			GPU latency (ms)		
Batch	T-128	S-32	S-16	T-128	S-32	S-16
1	20.82	6.86	6.11	46.84	43.64	26.14
8	16.07	2.95	2.14	10.37	9.02	6.89
32	27.05	4.68	2.35	4.02	2.68	3.08
64	27.85	5.85	2.87	3.44	1.85	1.30
128	—	—	—	1.54	1.56	1.42

Bottom line: “55× smaller” is a memory claim, not a speed claim. The latency win is real only on CPU under modest batching.

A calibration-preserving distillation loss for mixture predictors

What the thesis delivers

- Diagnosis: index-aligned KD is ill-posed (identity optimal in 0% of scenes)
- A permutation-invariant, matching-free mixture objective (any K_S, K_T)
- v1 breaks calibration by scale shrinkage; v2 = forward mode-covering KL fixes it
- ≈84% / 81% of a size-class recovered at zero inference cost; smaller students gain more

Limitations

- One dataset (Argoverse 1), one backbone (HiVT), one teacher
- HiVT-32 A/Bs are single-seed; Hungarian not tested empirically
- Hyperparameters ranked on a 25% proxy

Future work

- Other backbones / Argoverse 2, nuScenes; asymmetric $K_S \neq K_T$
- KD for learning longer trajectories (more time steps)

Ευχαριστώ για την προσοχή σας!